

CS 522: Distributed Algorithms

Course Code	CS 522
Course Name	Distributed Algorithms
L-T-P-C	3-0-0-3
Prerequisites	CS 202 Data Structures & Algorithms, CS 208 Mathematical Foundations of Computer Science or Equivalent or COT
Intended for	UG/PG
Distribution	Discipline Elective for BTech CSE, Free Electives for others
Approval	15th BoA (Dec 2016); 12th Senate (Mar 2017)

Course Contents

- **Module I:** Introduction (algorithmic challenges of distributed algorithms). [1 Lecture]
- **Module II:** Basic definitions (System models, distributed computation, local/global states, consistency, complexity measures). [6 Lectures]
- **Module III:** Time in distributed computing (logical clocks, vector clocks, virtual time). [3 Lectures]
- **Module IV:** Distributed graph algorithms (Spanning trees, broadcast & converge cast, shortest path). [6 Lectures]
- **Module V:** Distributed mutual exclusion algorithms. [4 Lectures]
- **Module VI:** Global state and snapshot recording algorithms (Chandy/Lamport , Lai/Yang, Mattern, Venkatesan Algorithm). [4 Lectures]
- **Module VII:** Monitoring global states (Necessary & sufficient conditions for consistent global states, zig-zag paths). [3 Lectures]
- **Module VIII:** Termination Detection (Based on snapshots computations, weight throwing, spanning trees). [3 Lectures]
- **Module IX:** Deadlock detection. [2 Lectures]
- **Module X:** Global predicate detection. [2 Lectures]
- **Module XI:** Checkpointing and rollback recovery (Coordinated, communication induced, ...). [4 Lectures]
- **Module XII:** Consensus and agreement algorithms. [4 Lectures]

Text Books

1. A. Kshenkalyani, M.Singhal, **Distributed Computing**, Cambridge University Press, 2008

References

1. Wan Fokkink, **Distributed Algorithms: an Intuitive Approach**, MIT Press, 2013
2. David Peled, **Distributed Computing: a Locality Sensitive Approach**, SIAM Monograph, 2000
3. Nancy Lynch, **Distributed Algorithms**, Morgan Kaufmann, 1996
4. M. Raynals, **Distributed Algorithms for Message-Passing Systems**, Springer, 2013